

figure #1

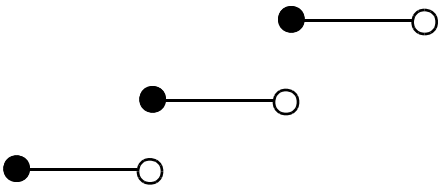
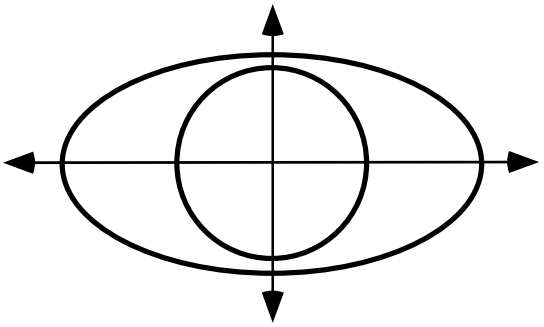
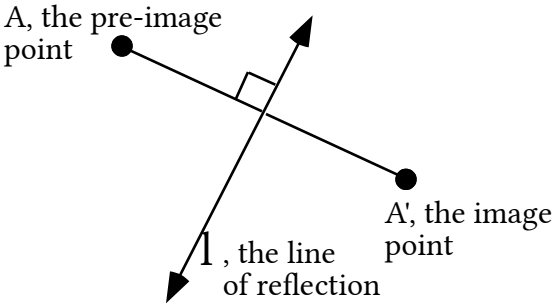
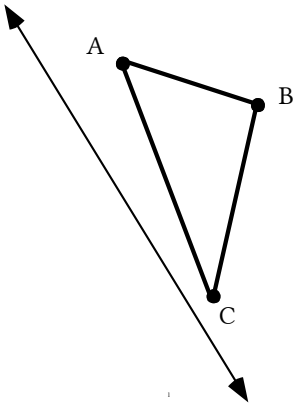
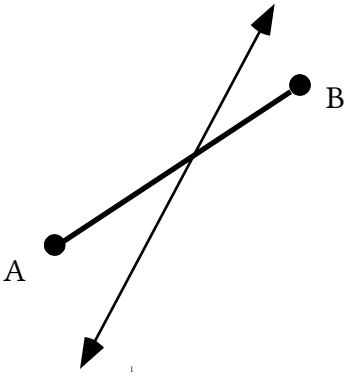


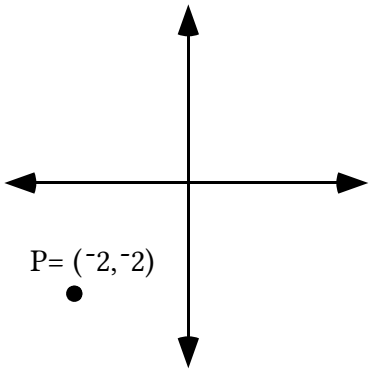
figure #2

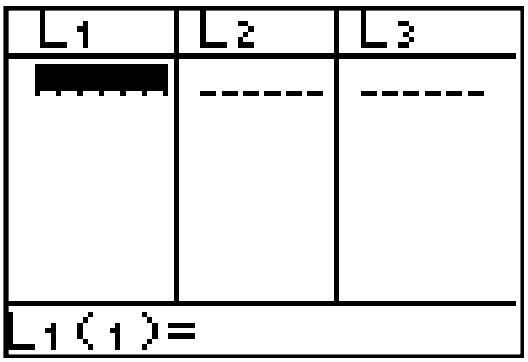



A, the pre-image point
A', the image point
l , the line of reflection


A
B
C


B
A


P= (-2,-2)


L1
L2
L3
L1(1)=

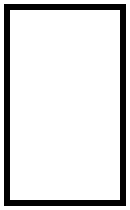


figure A

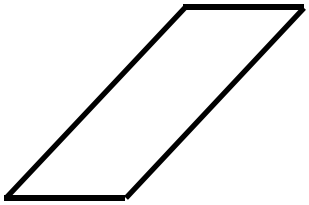
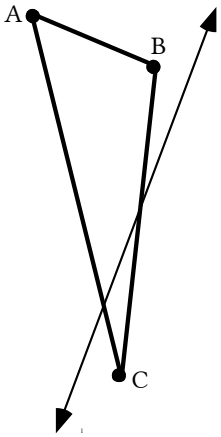
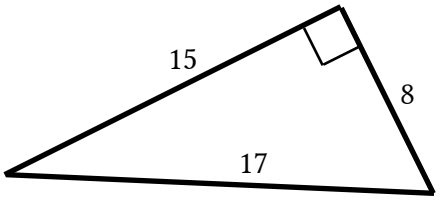
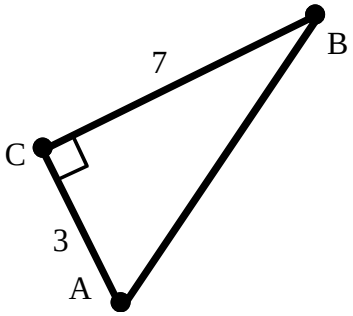


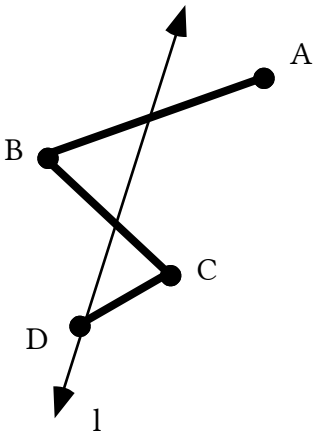
figure B

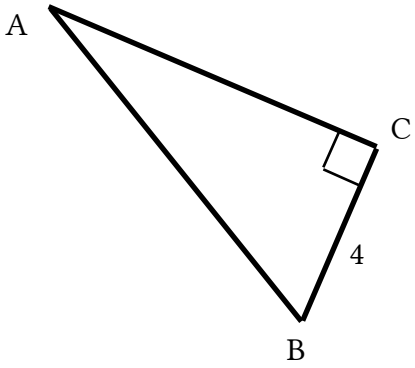

A
B
C

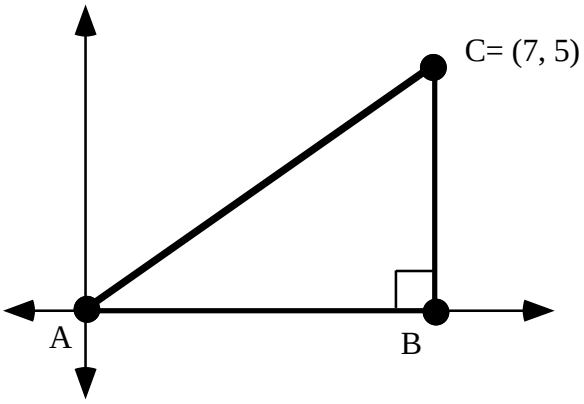

15
8
17

$$a^2 + b^2 = c^2$$

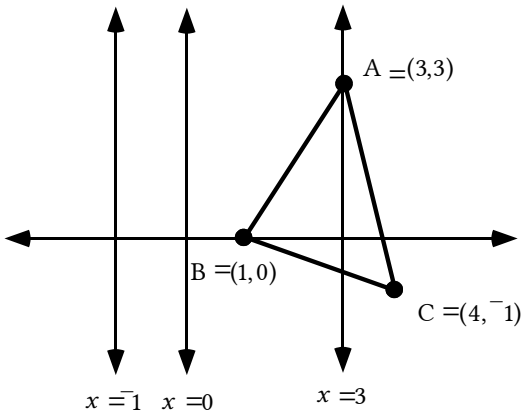

B
7
C
3
A

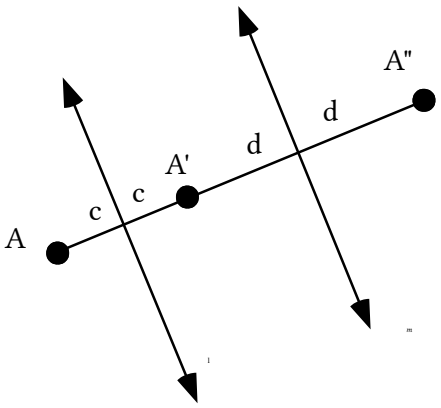

A
B
C
D
l

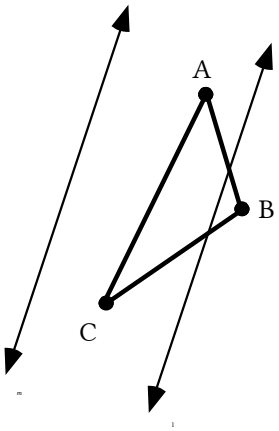

A
C
4
B

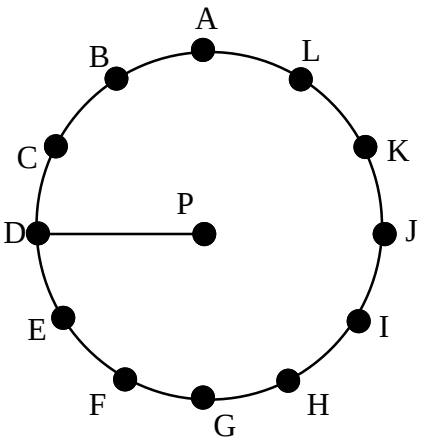

C= (7, 5)
A
B

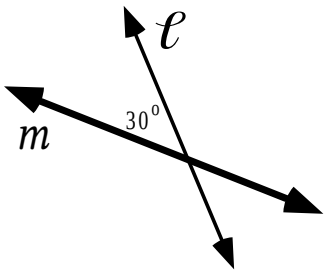
$r_{x=2}$ o $r_{x=-3}$

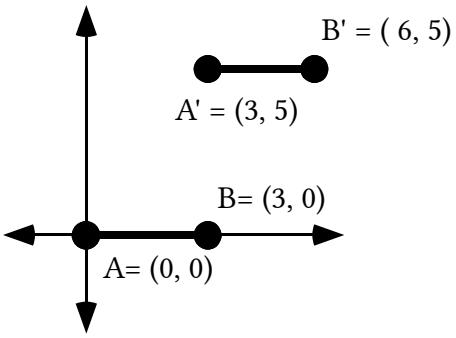

A =(3,3)
B =(1,0)
C =(4,−1)
x =−1
x =0
x =3

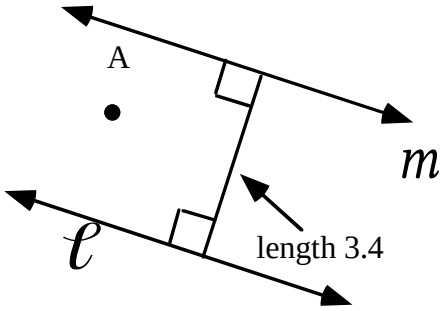

A"
d
d
A'
c
c
A

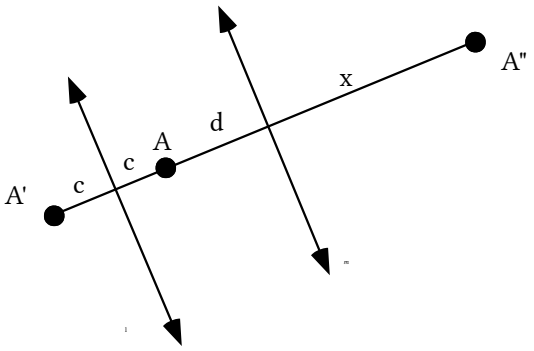

A
B
C

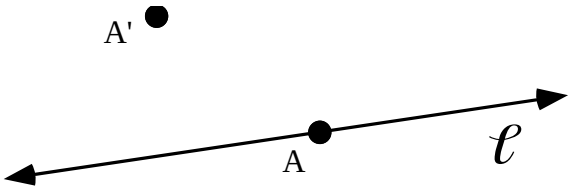

A
B
L
C
K
P
D
J
E
I
F
G
H

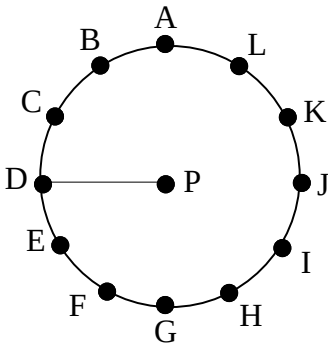

ℓ
30°
m

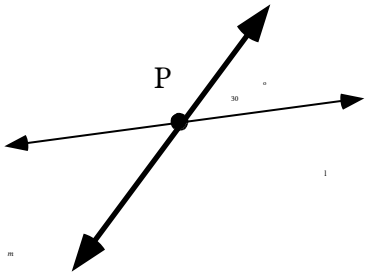

B' = (6, 5)
A' = (3, 5)
B= (3, 0)
A= (0, 0)

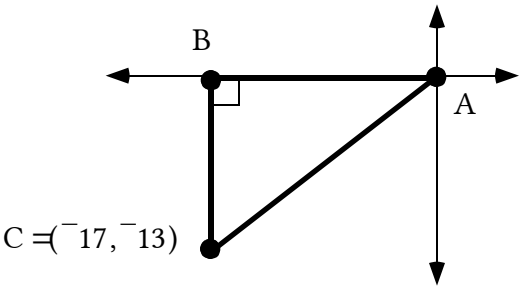

A
m
ℓ
length 3.4

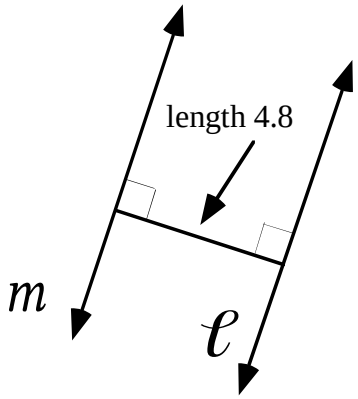

A''
x
d
A
c
c
A'

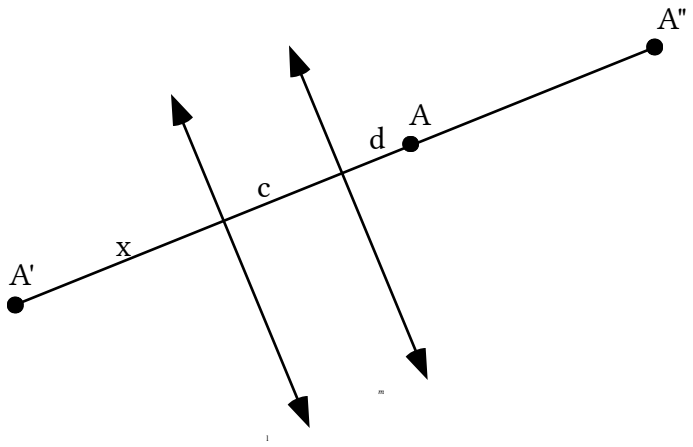

A'
A
ℓ

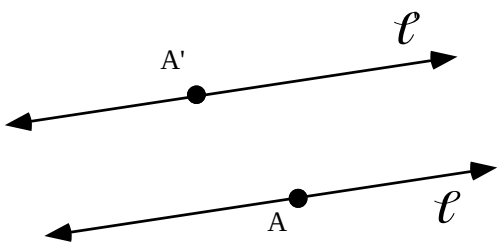

A
B
L
C
K
D
P
J
E
I
F
H
G

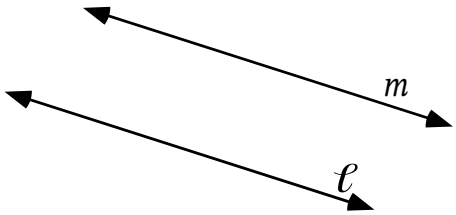

P
30
°
l
m

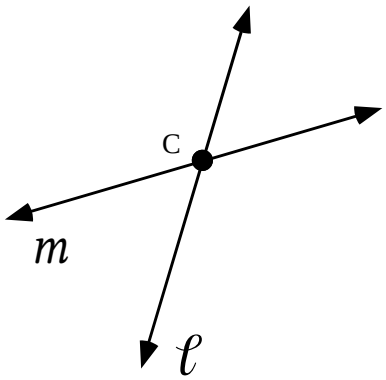

B
A
C = (−17, −13)

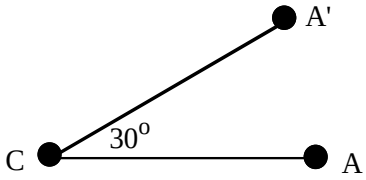

length 4.8
m
ℓ


A"
A
d
c
x
A'


A'
A

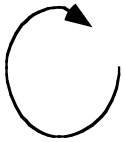

m
ℓ


C
m
ℓ

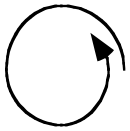




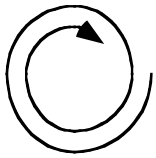
30 degrees
counter-clockwise



330 degrees
clockwise

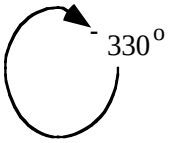


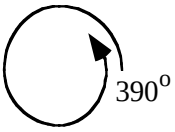
390 degrees
counter clockwise

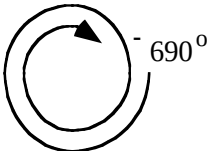


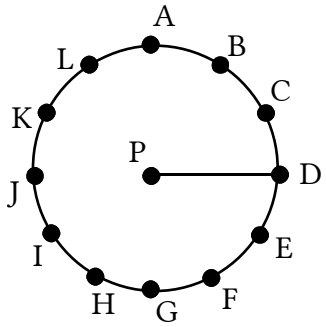
690 degrees
clockwise

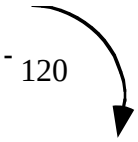




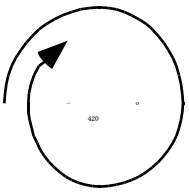


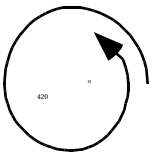


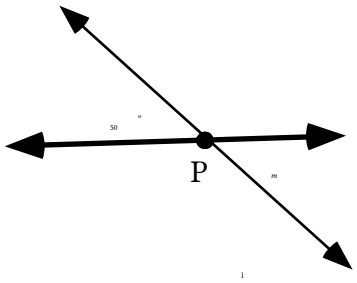

A
B
C
D
E
F
G
H
I
J
K
L
P

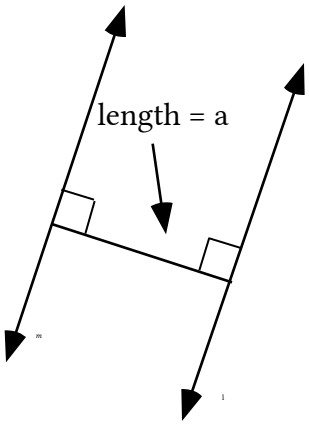

120

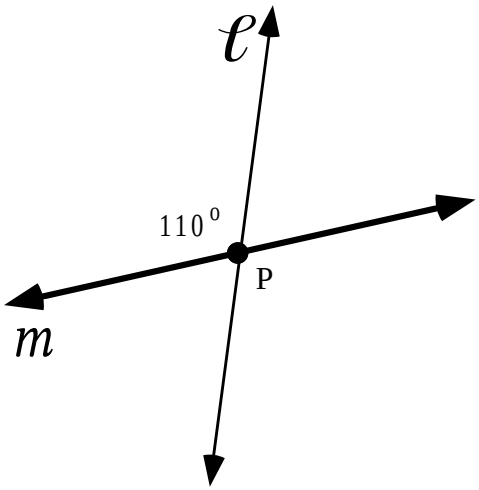
240

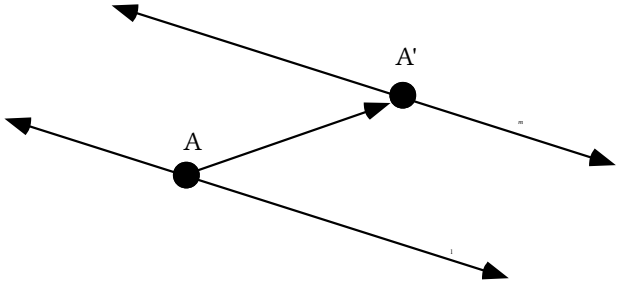

420

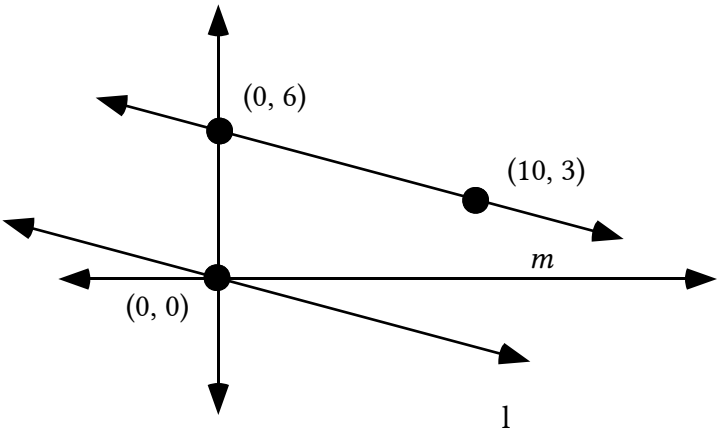

420

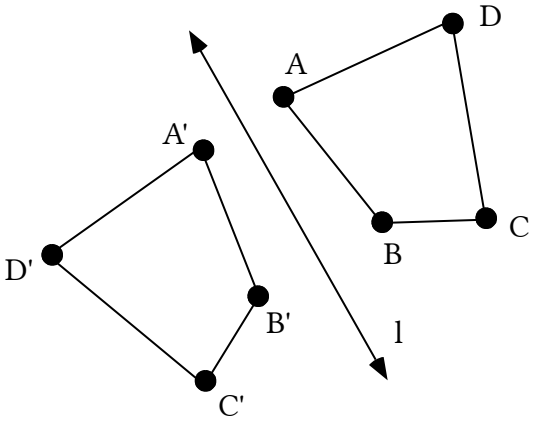

o
50
P
m
l

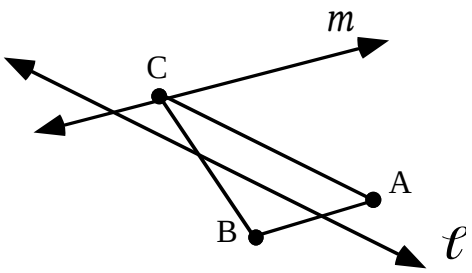

length = a

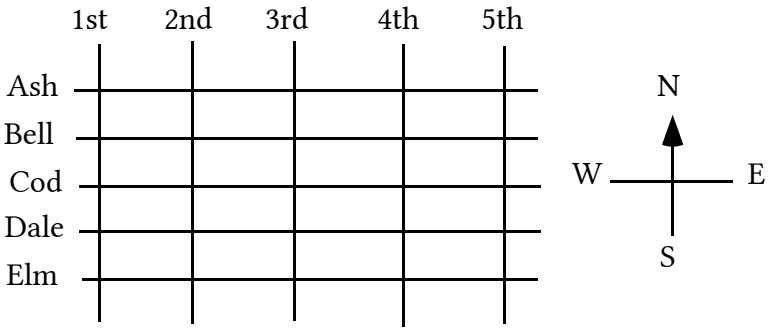

ℓ
110°
P
m

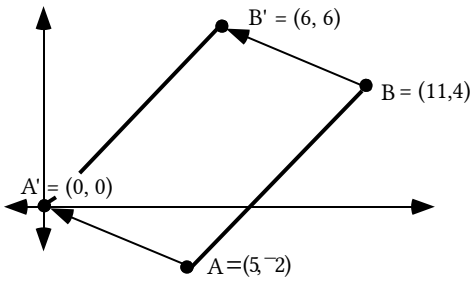

A'
A

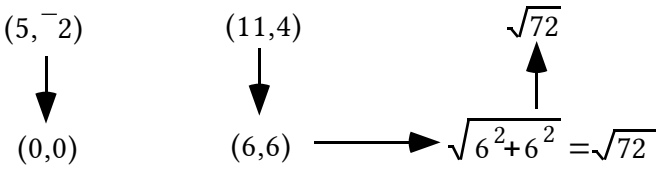

(0, 6)
(10, 3)
m
(0, 0)
l

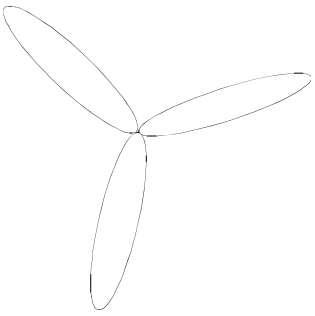

D
A
A'
C
B
D'
B'
l
C'

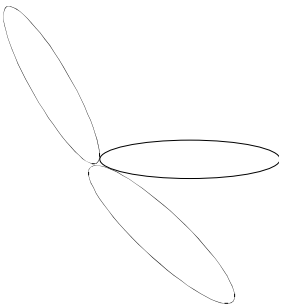

m
C
A
B
ℓ

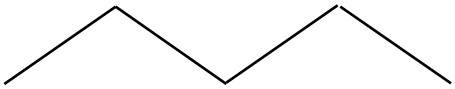

1st
2nd
3rd
4th
5th
Ash
Bell
Cod
Dale
Elm
N
W
E
S

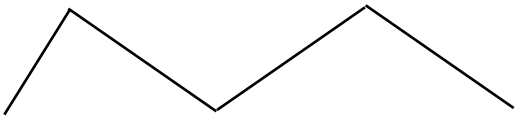

B' = (6, 6)
B = (11,4)
A' = (0, 0)
A =(5, −2)

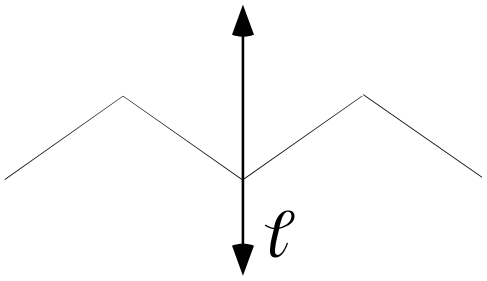

(5, ¯2)
(11,4)
√72
(0,0)
(6,6)
√6 2+6 2 =√72

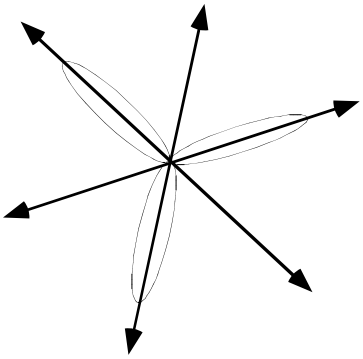


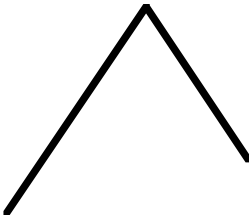


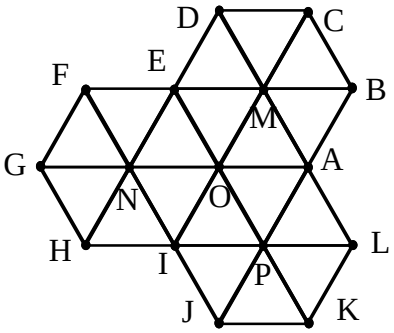


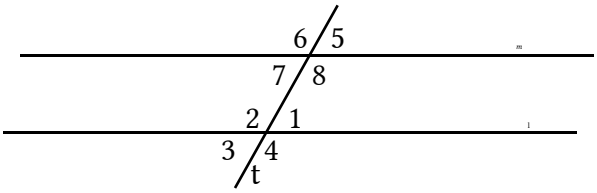



ℓ



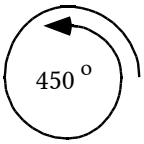


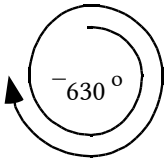

D
C
E
F
B
M
G
A
N
O
H
L
I
P
J
K

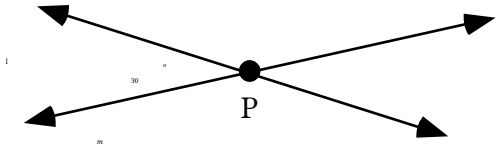

6
5
7
8
2
1
3
4
t

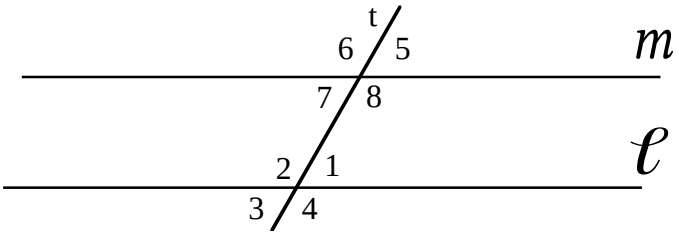
$r_x \circ r_{y=-x}$

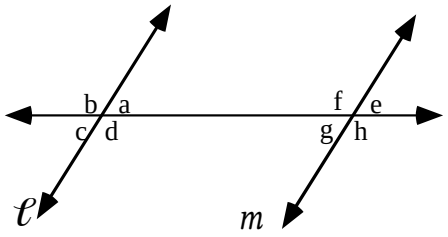

90 o
−270 o

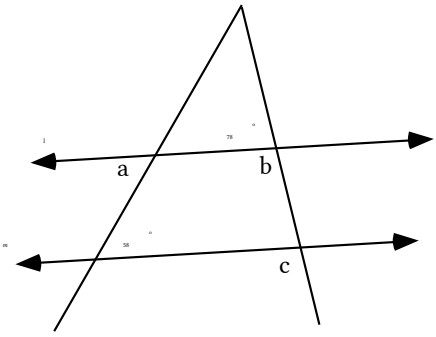

450 o

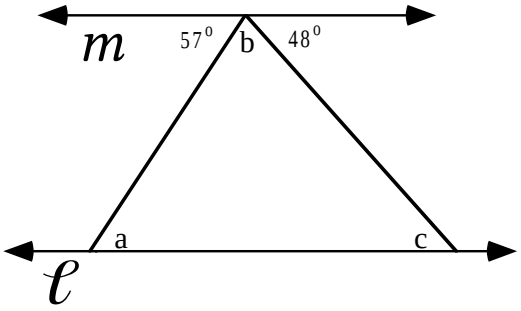

−630 o

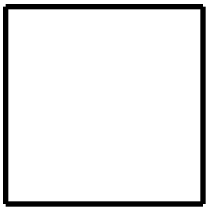

l
o
30
P
m

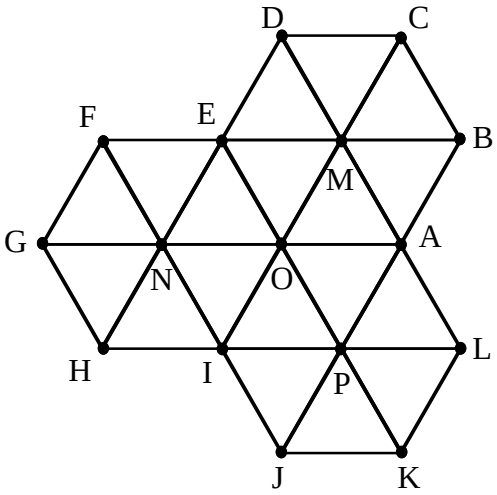

t
6
5
m
7
8
2
1
ℓ
3
4

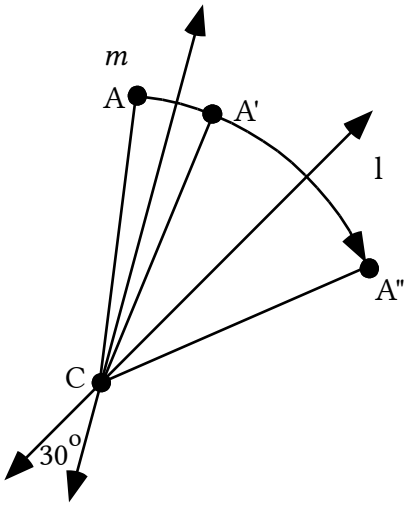

b
a
f
e
c
d
g
h
ℓ
m

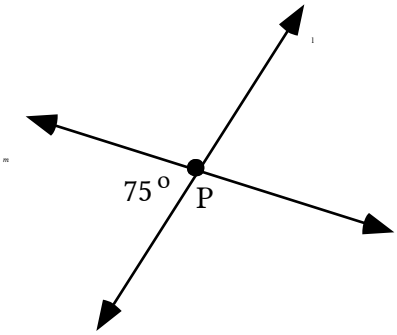

a
b
c

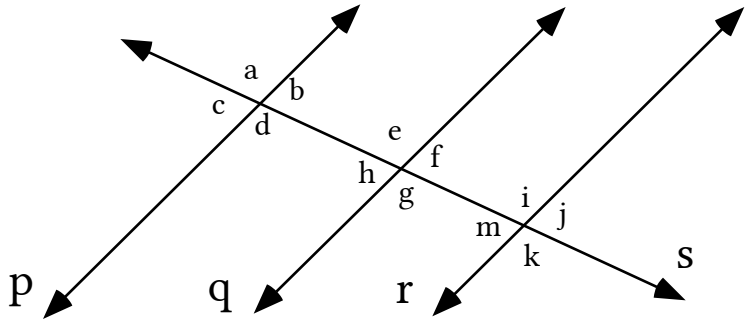

m
57°
b
48°
a
c
ℓ

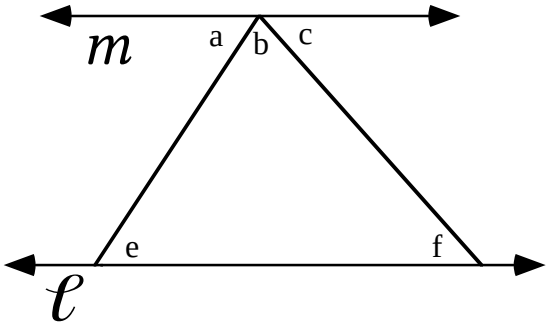


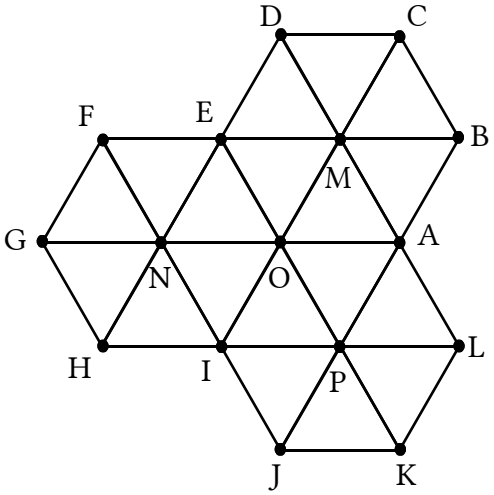

D
C
F
E
B
M
G
A
N
O
L
H
I
P
J
K

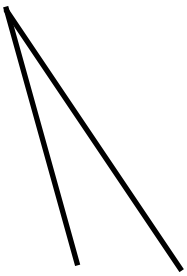

m
A
A'
l
A"
C
30°


75°
P

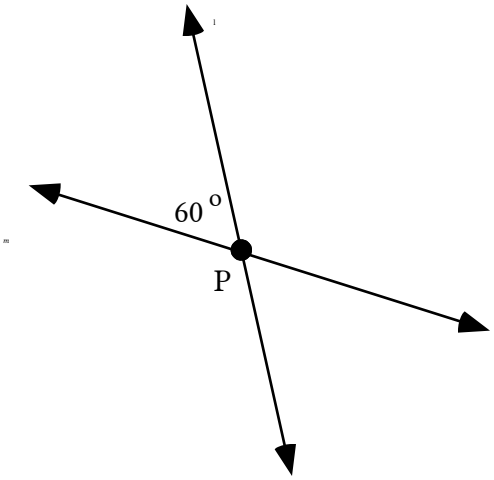

a
b
c
d
e
f
h
g
i
j
m
k
p
q
r
s

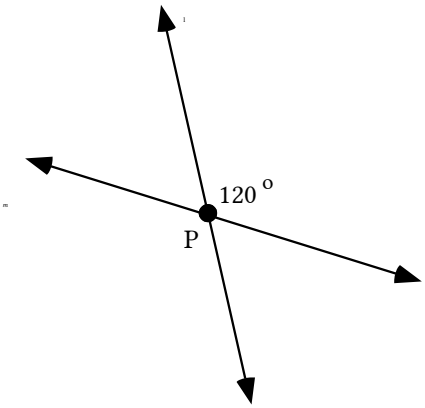

m
a
b
c
e
f
ℓ

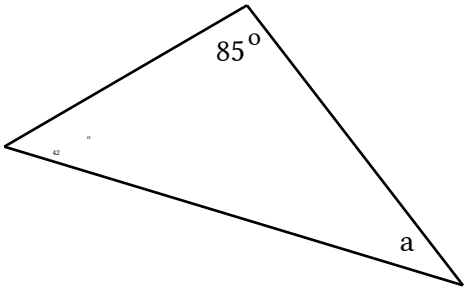

D
C
F
E
B
M
G
A
N
O
H
I
P
L
J
K

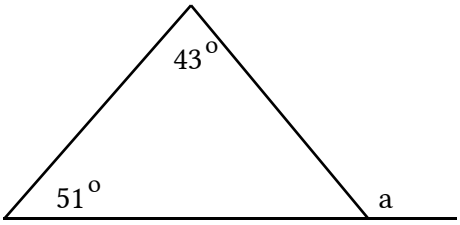


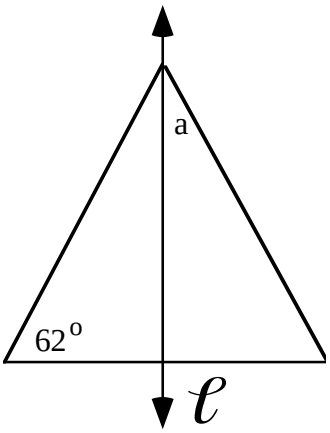


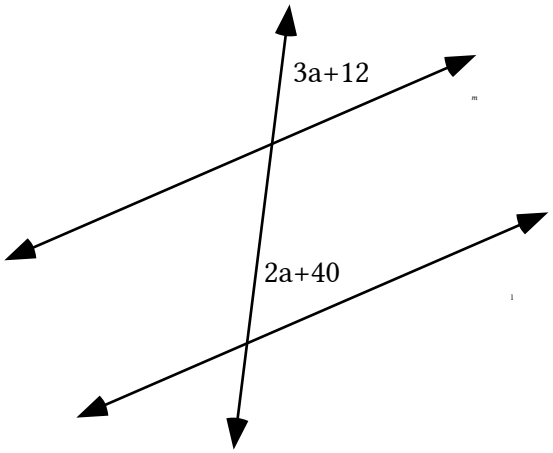

60 o
P

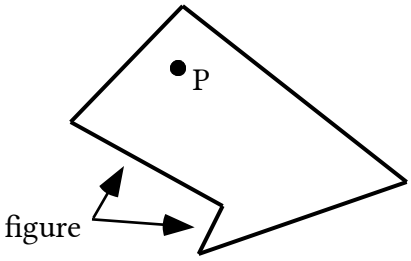

120 o
P

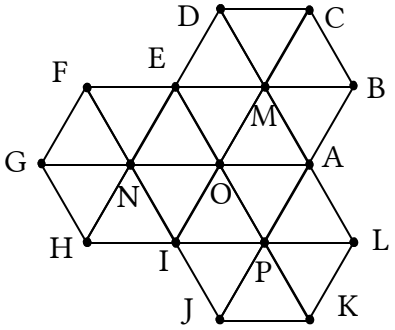

85°
42°
a


43°
51°
a


a
62°
ℓ


3a+12
2a+40


P
figure

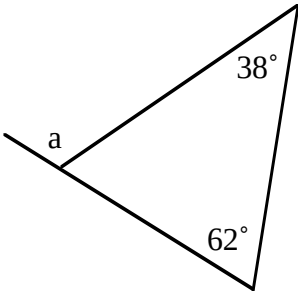

D
C
E
F
B
M
G
A
N
O
H
L
I
P
J
K

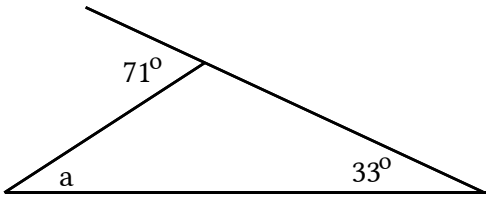
Matrix A -daily production
number of each style made

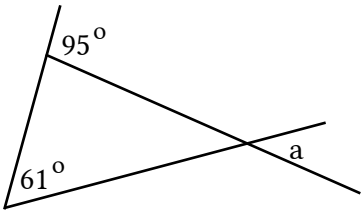
worker	style number #1	#2	#3
#1			
	[8	4	6]
#2			
	[6	6	8]

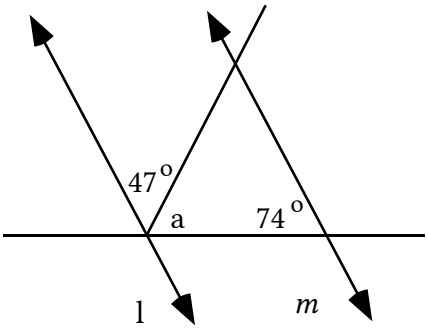
Matrix B -number of strips
of each type of wood
for each style

style	oak #1	walnut #2	cherry #3
#1	10	10	0
#2	8	6	6
#3	0	10	10


38°
a
62°


71°
a
33°


95°
a
61°


47°
a
74°
l
m

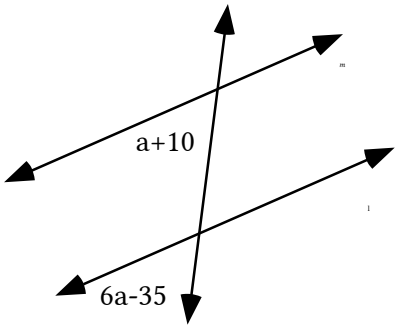


figure #1

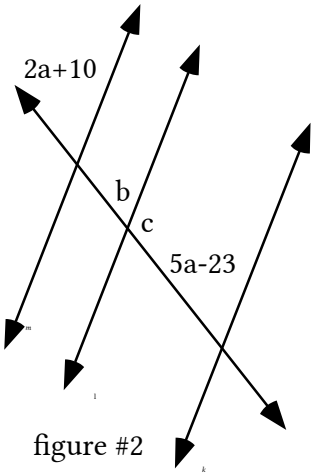
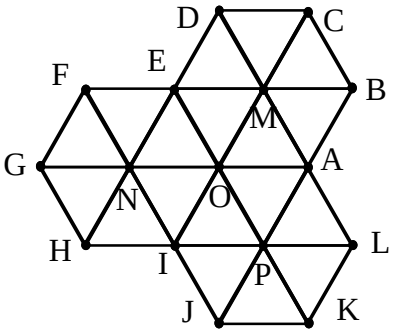
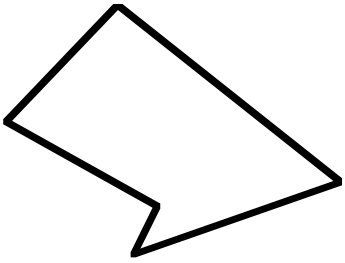
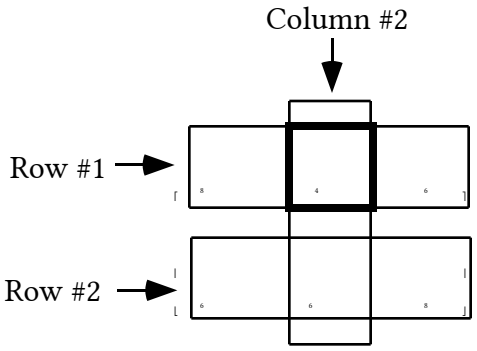


figure #2


D
C
E
F
B
M
G
A
N
O
H
L
I
P
J
K




Column #2
Row #1
Row #2

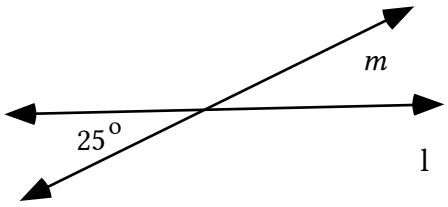
$$a_{21} \cdot b_{12} + a_{22} \cdot b_{22}$$

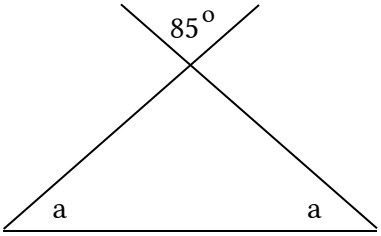
Matrix A -daily production number of each style made

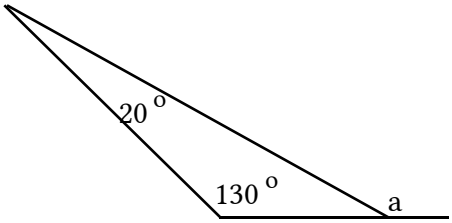
worker	style #1	style #2	style #3
#1	8	4	6
#2	6	6	8

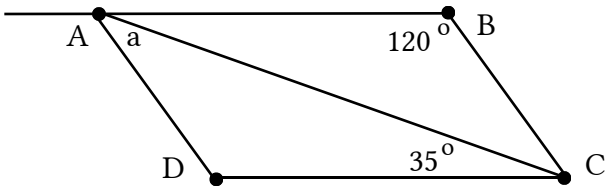
Matrix B -number of strips of each type of wood for each style

style	oak	walnut	cherry
#1	10	10	0
#2	8	6	6
#3	0	10	10


m
25°
l


85°
a
a


20 o
130 o
a


A
a
120 o
B
D
35 o
C

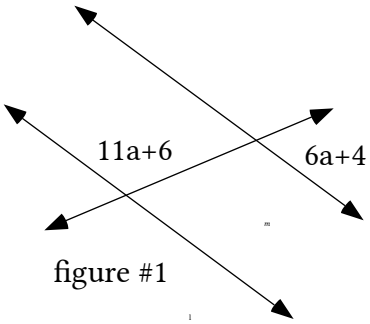


figure #1

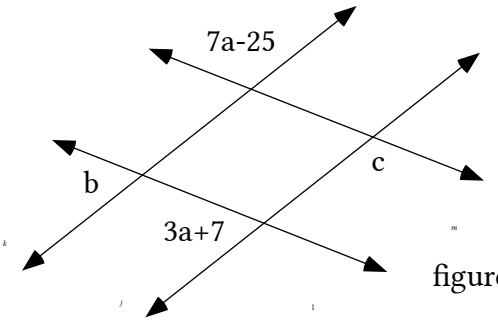


figure #2

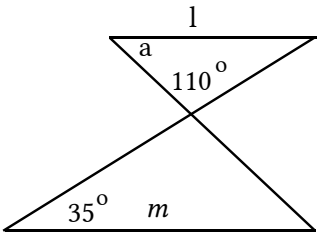
Matrix A -daily production number of each style made

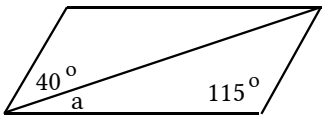
style number

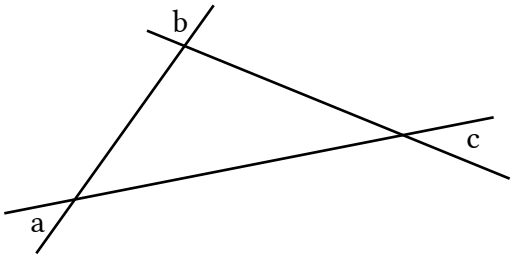
worker	#1	#2	#3
#1			
#2			

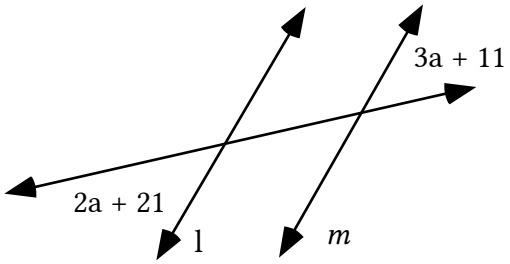
Matrix B -number of strips of each type of wood for each style

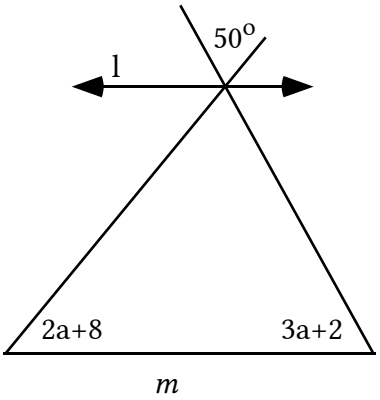
style	oak #1	walnut #2	cherry #3
#1	10	10	0
#2	8	6	6
#3	0	10	10

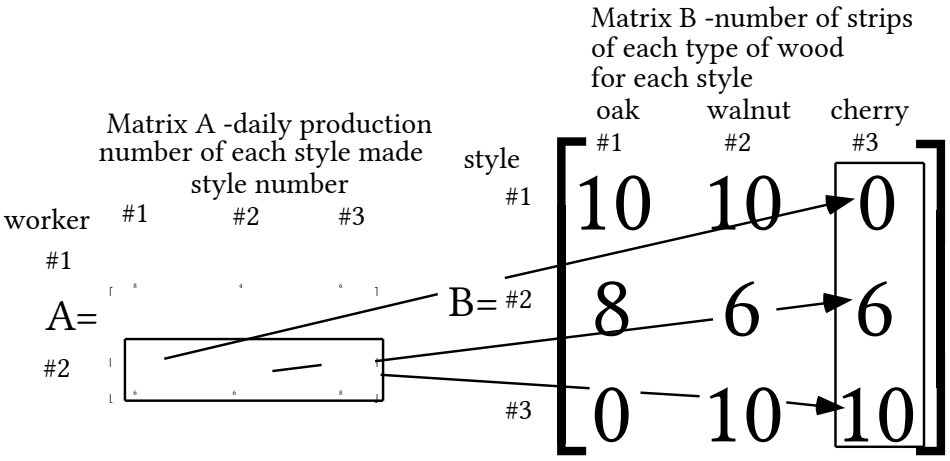

l
a
110 o
35 o
m


40 o
a
115 o


b
c
a


3a + 11
2a + 21
l
m


50°
l
2a+8
3a+2
m

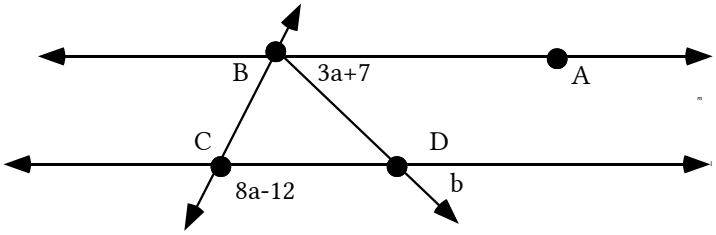

Matrix B -number of strips
of each type of wood
for each style
oak
walnut
cherry
#1
#2
#3
Matrix A -daily production
number of each style made
style number
worker
#1
#2
#3
A=
#1
#2
B=
#2
#3
10
10
0
8
6
6
0
10
10

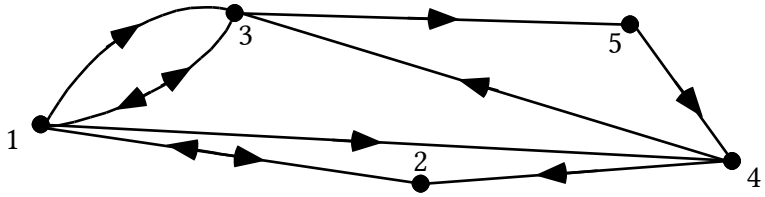
total number of strips needed each day

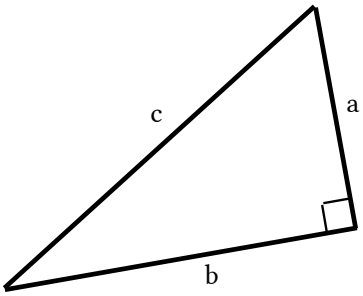
	oak	walnut	cherry
worker #1			
worker #2			116

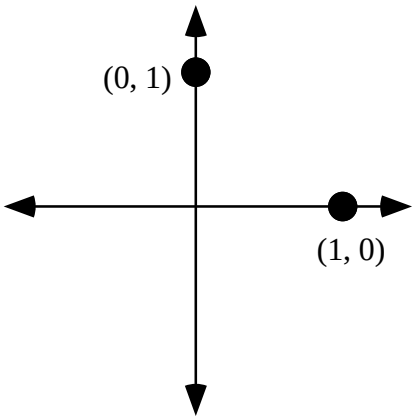
C=

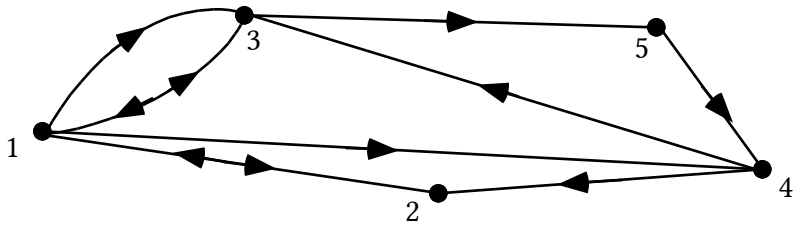
	Children	Students	Adults
Cinema I	225	110	50
Cinema II	75	180	225
Cinema III	280	85	110
Cinema IV	0	250	225

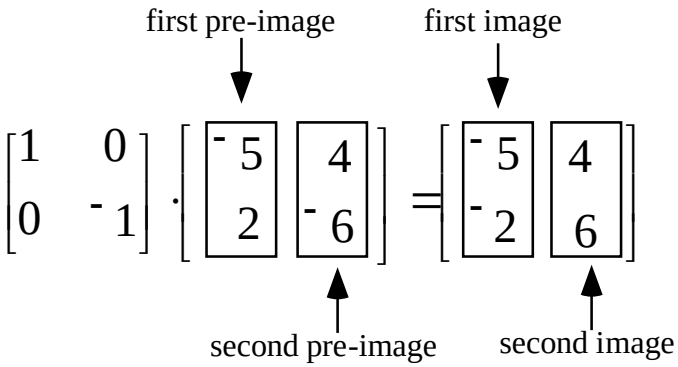

B
3a+7
A
C
D
8a-12
b

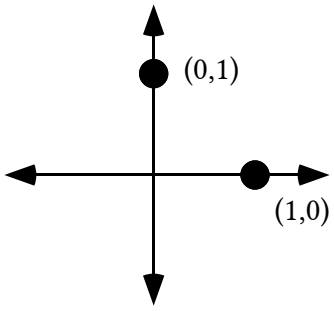

3
5
1
2
4

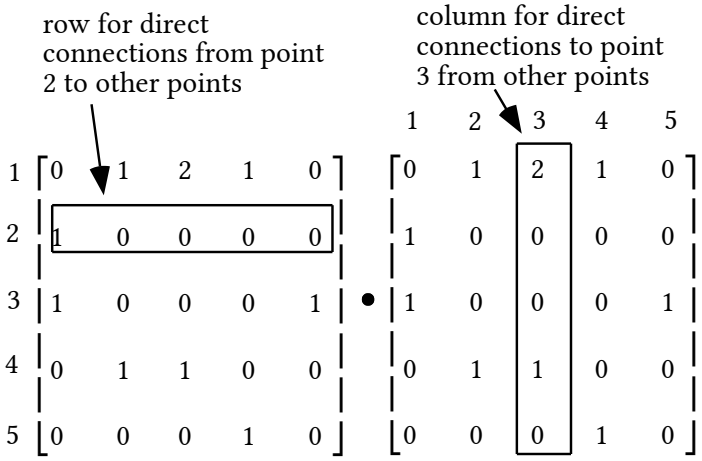

c
a
b

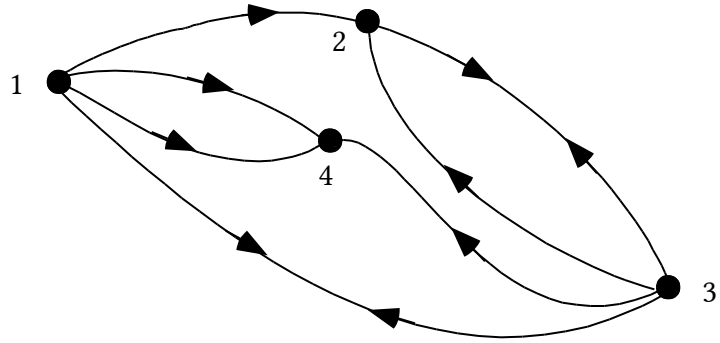

(0, 1)
(1, 0)

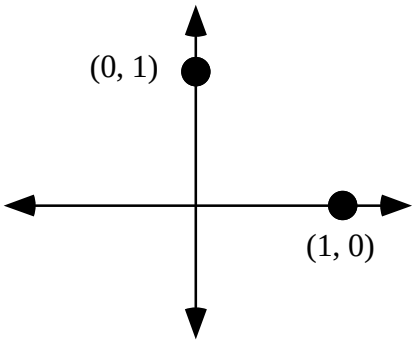

3
5
1
4
2

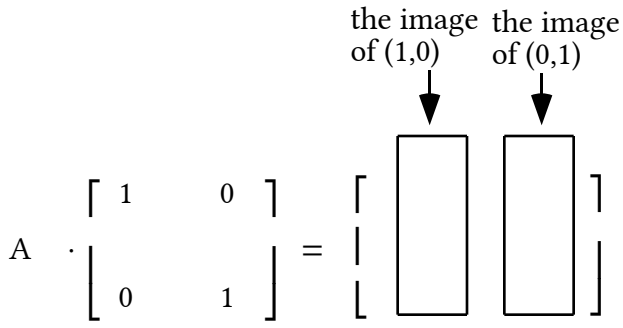

first pre-image
first image
second pre-image
second image

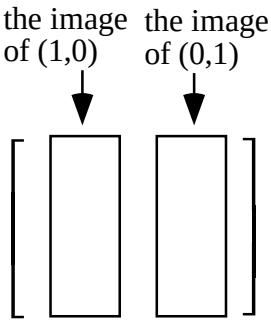

(0,1)
(1,0)

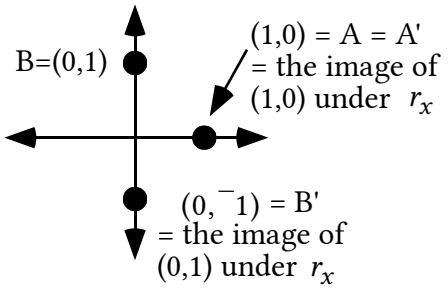

row for direct
connections from point
2 to other points
column for direct
connections to point
3 from other points
1 2 3 4 5
1 0 1 2 1 0
2 1 0 0 0 0
3 1 0 0 0 1
4 0 1 1 0 0
5 0 0 0 1 0
0 1 2 1 0
1 0 0 0 0
1 0 0 0 1
0 1 1 0 0
0 0 0 1 0

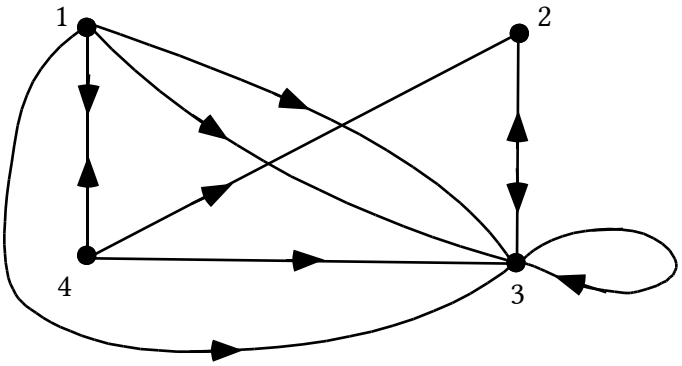

1
2
3
4

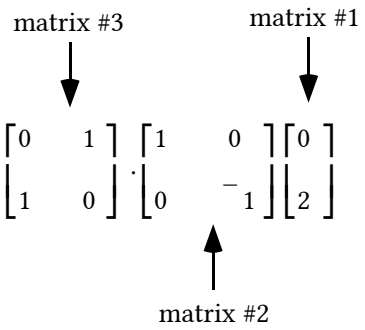

(0, 1)
(1, 0)


the image of (1,0)
the image of (0,1)
A
1
0
0
1
=


the image
of (1,0)
the image
of (0,1)


B=(0,1)
(1,0) = A = A'
= the image of
(1,0) under r_x
(0,¯1) = B'
= the image of
(0,1) under r_x


1
2
4
3


matrix #3
matrix #1
matrix #2